MIL-I-28775(Wep)
25 June 1963

SUPERSEDING
NAVORD OS 8160A
3 January 1963

MILITARY SPECIFICATION
INERT LOAD BINDER FOR DRILL MINES

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of Inert Load Binder for Drill Mines, for uses in applications wherein low magnetic permeability characteristics are required.

1.2 Classification - The Inert Load Binder for Drill Mines shall be of the following types:

Type I. - For use in general binding
Type IA. - Air-entraining binder for same uses as Type I.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this document.

SPECIFICATIONS

MILITARY

MIL-B-1190	Bags, Paper, Shipping, Multiwall (For Portland Cement)

FSC 5610

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for its intended purpose. Minor characteristics are characteristics whose nonconfor- item

STANDARDS

FEDERAL

Federal Test Method Standard No. 158	Cements, Hydraulic, Sampling, Inspection and Testing

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 10511	Basic Methods for Nonmagnetic Testing with Test Set Mark 115 Type

(Copies of specifications, standards, and publications required by contractors in accordance with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Designation C-115-42	Fineness of Portland Cement by the Turbidimeter

(Copies of ASTM Standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Description. - This specification describes Inert Load Binder for Drill Mines conforming to requirements and characteristics

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as detailed in 3.2 and 3.3. The binder shall be classed as Type I for use in general binding and Type IA air-entrained binder for use in general binding.

3.2 General requirements

3.2.1 Material. - The binder may be made from any selection of raw materials and by any process that will yield a product conforming to the requirements detailed in this specification.

3.2.2 Conflicting requirements. - Conflicting requirements arising between this specification or any specification or publication listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Performance requirements and product characteristics. The binder shall meet the following performance requirements and product characteristics:

3.3.1 Chemical requirements. - The binder shall conform to the chemical composition indicated in Table I.

3.3.2 Physical requirements. - The binder shall conform to the physical requirements indicated in Table II.

3.3.3 Static magnetic effect. - When tested in accordance with paragraph 4.5.5, the static magnetic effect of the material as tested shall not exceed 0.10 milligauss.

4. QUALITY ASSURANCE PROVISIONS

4.1 Performance of sampling inspection and testing. - Unless otherwise specified, sampling, inspection and testing for acceptance of individual lots shall be performed by the contractor. Contractors not having laboratory facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring activity. Detail records of all contractor performed examination and tests shall be made available to the Government representative.

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4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provision set forth in MIL-STD-105.

4.1.2 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean "Inspection lot", and shall consist of one batch offered for delivery at one time.

4.2 Lot acceptance tests. - The sample selected in accordance with 4.1.1 shall be subjected to the tests specified in 4.5. When samples are submitted to represent more than one lot under a contract or order, the laboratory may elect to perform complete tests on one sample and only selective tests on other samples submitted to specify that all material is alike.

4.3 Rejected lots. - Rejected lots may be resubmitted for inspection and acceptance tests provided the contractor has removed (or reworked) all nonconforming products.

4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.1 Standard commercial testing equipment for obtaining chemical analysis and physical characteristics of the binder material.

Test Set Mark 115 Mod 4	Government Furnished Equipment

4.4.2 The Government shall furnish Test Set Mark 115 Mod 4; the Contractor shall furnish personnel, facilities and other equipment to accomplish the acceptance tests.

4.5 Acceptance tests.

4.5.1 Chemical analysis. - Federal Test Method Standard No. 158, Cements, Hydraulic; Sampling, Inspection, and Testing shall be used for obtaining chemical analysis. Referee methods shall be used in case of dispute, or when the results obtained by optional methods indicate a result so close to the specification limit that meeting or failing to meet the specification requirement is in doubt. When referee

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methods are required all determinations of constituants involved in the computations shall be made entirely by referee methods and the separation prior to these methods shall be made by referee methods.

4.5.2 Fineness tests. - The fineness test utilizing the Blaine fineness meter shall be in accordance with Federal Test Method Standard No. 158, Cements, Hydraulic; Sampling, Inspection, and Testing. The fineness test utilizing the Wagner turbidimeter shall be performed in accordance with ASTM Designation C-115-42, Fineness of Portland Cement by the Turbidimeter.

4.5.3 Soundness. - Binder failing to conform to the test for soundness indicated in Federal Test Method Standard No. 158, Cements, Hydraulic; Sampling, Inspection, and Testing may be accepted if it passes in a retest of a new sample taken within 28 days of the time of the original sampling. The provisional acceptance of the binder at the mill shall not deprive the purchaser of the right of rejection on a retest of soundness at the time of the delivery of the binder to the purchaser.

4.5.4 Time of setting tests. - Determination of the time of setting shall be performed in accordance with Federal Test Method Standard No. 158, Cements, Hydraulic; Sampling, Inspection, and Testing.

4.5.5 Static magnetic effect test. - The magnetic testing of binder material shall be conducted with the Government furnished Test Set Mark 115 Mod 4. The binder shall be tested in accordance with Test Method I of OD 10511, under the general test conditions and definitions described therein, except that only Orientation (A) need be used. The test quantity shall be 10 pounds of binder selected to fill a box-shape nonmagnetic test container. The base of the container shall be square and shall be the test face. The height of the container shall equal approximately one-half the length. The test distance shall be 1 3/4 inches from the center of the test magnetometer to the test face. When tested as prescribed, binder material producing a magnetic signal exceeding 0.10 milligauss, measured as described in OD 10511, shall be considered as defective and shall be the basis for rejection of the batch or lot offered for shipment.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - (Not applicable)

5.2 Packing. - Unless otherwise specified, binder material shall be shipped in bags, as specified. Bags differing by more than 5 percent from the specified weight may be rejected; and if the average

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weight of 50 packages taken at random in any shipment is less than that specified, the entire shipment may be rejected. When shipped in bags, packing shall be in accordance with one of the following levels, as specified.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Binder, in quantities of 94 pounds net shall be packed in Grade II, Types C, D or E multiwall bags, conforming to the requirements of Military Specification MIL-B-1190. Selection of bag shall be as specified by the procuring activity for Level A packing requirements.

5.2.2 Level B.

5.2.3 Level C.

5.2.3.1 Exterior containers. - Binder, in quantities of 94 pounds net shall be packed in Grade I, Types A or B multiwall bag conforming to the requirements of Military Specification MIL-B-1190. Selection of bag shall be as specified by the procuring activity.

5.3 Marking.

5.3.1 Normal markings. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Specification MIL-STD- 129.

5.3.2 Special marking. - Special markings shall be in accordance with Military Specification MIL-B-1190.

6. NOTES

6.1 Intended use. - The binder is intended for initial use in drill mines.

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6.2 Ordering data. - Procurement documents should specify the desired level of packaging and packing. Level C should not be selected unless it is definitely known that the material will be used immediately upon receipt.

6.3 The manufacturers listed in this paragraph have supplied inert load binder which has been tested in accordance with this specification and proved to be satisfactory. Therefore, if a contract is awarded to any of these manufacturers, samples shall be tested in accordance with procedures of reduced testing as indicated in Federal Test Method Standard No. 158.

MANUFACTURER	TYPE
Medusa Portland Cement Co. York, Pa.	I
General Portland Cement Co. 111 W. Monroe Street Chicago 3, Ill.	I

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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TABLE I - Chemical Requirements[1]

	TYPES I	TYPES IA
Silicon dioxide (SiO_2)	- - -	- - -
Aluminum oxide (Al_2O_3) maximum	7.5	7.5
Ferric oxide (Fe_2O_3) maximum	0.4	0.4
Magnesium oxide (MgO) maximum	5.0	5.0
Sulfur trioxide (SO_3) maximum	3.0	3.0
Loss on ignition, maximum	3.0	3.0
Insoluble residue, maximum	0.75	0.75
Tricalcium silicate (C_3S = 3CaO. SiO_2) [2], average	41.2 - 51.2	49.1
Dicalcium silicate (C_2S = 2CaO. SiO_2) [2], average	25.7 - 41.8	28.8
Tricalcium aluminate (C_3A = 3CaO. Al_2O_3) [2], average	11.2 - 14.6	14.5
Dicalcium silicate (C_2S = 2CaO. SiO_2) [2], average	25.7 - 41.8	28.8
Tricalcium aluminate (C_3A = 3CaO. Al_2O_3) [2], average	11.2 - 14.6	14.5
Tetracalcium aluminoferrite (C_4AF = 4CaO. Fe_2O_3) [2], average	0.8 - 1.0	1.0
Free lime, average	1.79	1.52

[1] Referee methods shall be used in case of dispute, or when the results obtained by optional methods indicate a result so close to the specification limit that meeting or failing to meet the specification requirements is in doubt. When referee methods are required, all determinations of constituants involved in the computation, shall be made entirely by referee methods, and the separation prior to these determinations shall be made by referee methods.

[2] The expressing of chemical limitations by means of calculated assumed components does not necessarily mean that the oxides are actually or entirely present as such components. The percentages of tricalcium silicate, dicalcium silicate, tricalcium aluminate and tetracalcium aluminoferrite, shall be calculated from the chemical analysis as follows:

Tricalcium silicate = (4.07 x percent CaO) - (7.60 x percent SiO_2) - (6.72 x percent Al_2O_3) - (1.43 x percent Fe_2O_3) - (2.85 x percent SO_3)

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TABLE I. - Chemical Requirements [1] (Cont)

Dicalcium silicate = (2.87 x percent SiO_2) - (0.754 x percent 3 CaO. SiO_2)

Tricalcium aluminate = (2.65 x percent Al_2O_3) - (1.69 x percent Fe_2O_3)

Tetracalcium aluminoferrite = (3.04 x percent Fe_2O_3)

Oxide determinations calculated to the nearest 0.1 percent shall be used in the calculation. Compound percentages shall be calculated to the nearest 0.1 percent and reported to the nearest 0.1 percent.

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TABLE II. - Physical Requirements

	TYPES	
	I	IA
Fineness, Blaine fineness meter:		
Average value, minimum . . . . . sq. cm./g	3700	3650
Fineness, Wagner turbidimeter		
Average value, minimum . . . . . sq. cm./g	1800	1785
Soundness, autoclave expansion, maximum . . percent	0.15	0.11
Time of setting. Gillmore test:		
Initial, minimum . . . . . minutes	180	180
Final, maximum . . . . . hours	6	6

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SPECIFICATION ANALYSIS SHEET
NAVMEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through the use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION (No. and short title)

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THIS SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

REMARKS (Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)

SUBMITTED BY (Print name and title)	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

Navy Department

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